

News, Price Volatility and Market Activity in Major Cryptocurrencies.

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Abstract

This project investigates the relationship between news coverage, price volatility, and market activity in major cryptocurrency markets. Hourly market data for Bitcoin (BTC), Ethereum (ETH), Binance Coin (BNB), Solana (SOL), and Ripple (XRP) are collected from the Binance API and integrated with hourly news indicators from the GDELT API over a two-year period, resulting in an analysis dataset exceeding 80,000 asset-hour observations. After aligning both sources on a shared hourly timeline and storing the curated dataset in an SQL database, a set of time-window queries is used to evaluate the association between news dynamics and market behaviour, explicitly accounting for potential lagged effects. News spikes, defined using asset-specific thresholds computed within each semester to account for regime changes in media intensity are associated with higher subsequent price movement and trading activity. Conversely, elevated volatility often precedes news spikes, and extreme price moves are frequently followed by higher news volume and an increased probability of observing at least one news spike within the next 24 hours, supporting a bidirectional relationship between market shocks and media attention.

Contents

1. INTRODUCTION	2
1.1 Research Questions	2
2. DATA ACQUISITION	3
2.1 Collected Variables	3
3. PRE-INTEGRATION DATA QUALITY	4
4. DATA INTEGRATION.....	5
5. DATA ENRICHMENT	5
5.1Post Enrichment Data Quality.....	6
6. DATA STORAGE	6
7. DATA ANALYSIS.....	7

7.1 RQ1. News spikes and subsequent price movement	7
7.2 RQ2. News spikes and previous price movements	9
7.3 RQ3. News spikes and subsequent market activity.....	10
7.4 RQ4. Price spikes and subsequent news coverage	12
CONCLUSIONS.....	14
LIMITATIONS	14
FUTURE WORK	15
REFERENCES.....	15

1. Introduction

Cryptocurrency markets are highly dynamic and often react to new information within short time windows. Prices can move sharply due to a wide range of drivers, including macroeconomic announcements, regulatory decisions, exchange-specific events, and changes in investor sentiment. At the same time, news coverage itself may react to the market: large price movements can attract media attention and generate further reporting. For these reasons, understanding the timing of the interaction between news and prices is crucial, especially at an hourly frequency where lagged and non-immediate effects may emerge.

This project focuses on the relationship between news coverage, price volatility, and market activity in major cryptocurrencies. We study five assets Bitcoin (BTC), Ethereum (ETH), Binance Coin (BNB), Solana (SOL), and Ripple (XRP) combining hourly market data with hourly news signals. The central challenge is temporal: to evaluate whether communication impacts market behaviour (and vice versa), the two sources must be aligned on a common timeline and analyzed with explicit lead/lag horizons. To address this, we build a curated hourly dataset and use SQL-based analyses with time-window queries to compare market behaviour around news spikes and price shocks.

The contribution of the project is twofold. First, we deliver an end-to-end data management workflow, from API-based acquisition to a structured dataset suitable for analysis. Second, we provide empirical evidence on the bidirectional dynamics between news and market behaviour, by evaluating both directions: news spikes followed by price and activity changes, and price spikes followed by increased news coverage.

1.1 Research Questions

The project is guided by four research questions aimed at capturing the timing and direction of the relationship between news coverage and cryptocurrency market dynamics.

RQ1. News spikes and subsequent price movement

Do hours characterized by a spike in news coverage coincide with larger price movements in the following horizons: 1 hour, 6 hours, and 24 hours?

RQ2. Price movement preceding news spikes

Are news spikes preceded by higher price volatility in the previous horizons: 1 hour, 6 hours, and 24 hours, suggesting that news coverage may react to ongoing market moves?

RQ3. News spikes and subsequent market activity

Do news spikes correspond to increased market activity, measured through the number of trades, in the subsequent horizons: 1 hour, 6 hours, and 24 hours?

RQ4. Price spikes and subsequent news coverage

Do extreme price movements increase subsequent news coverage, measured both as future news volume and as the probability of observing at least one news spike within the next 24 hours?

2. Data Acquisition

Hourly datasets were collected for five major cryptocurrencies (BTC, ETH, BNB, SOL, XRP) by combining market data and news signals at 1-hour granularity over a two year period. Data collection covers the period starting on January 25, 2024 up to January 24, 2026, ensuring that both sources can be aligned on a consistent hourly timeline.

Market data were retrieved from the Binance Spot REST API using the *klines* endpoint with the interval set to 1 hour. Since each request returns a limited number of candles, the extraction was implemented through time-based pagination over the selected time range, and exported to CSV files. To improve robustness during acquisition, the downloader includes retry logic with short delays to handle transient failures and potential rate-limit responses, and all timestamps were standardized to UTC to support downstream integration.

News signals were collected from the GDELT DOC 2.0 API via the *timeline* interface, which provides aggregated hourly indicators without downloading full article lists. For each cryptocurrency, keyword-based queries were designed to capture articles explicitly related to the asset and the crypto domain, reducing false positives. To preserve a stable time granularity and avoid resolution changes for longer requests, the extraction was executed in weekly sub-windows. Requests were throttled to comply with API limits (approximately one request every five seconds), and backoff was applied in case of HTTP 429 responses. The resulting hourly series were exported and standardized to UTC to enable direct alignment with market data on an hourly key.

2.1 Collected Variables

Binance

- ◆ symbol: Trading pair identifier for the asset.
- ◆ open_time: Start timestamp of the 1-hour candle (opening time), used as the primary time reference for hourly alignment.
- ◆ close_time: End timestamp of the 1-hour candle (closing time).
- ◆ open: Opening price of the asset in the hour.
- ◆ high: Highest traded price within the hour.
- ◆ low: Lowest traded price within the hour.
- ◆ close: Closing price of the asset in the hour.
- ◆ n_trades: Number of executed trades during the hour, used as a proxy for market activity/intensity.

GDELT

- ◆ asset: Cryptocurrency identifier (BTC, ETH, BNB, SOL, XRP) associated with the query.

- ◆ **ts_hour_utc**: Hourly timestamp in UTC representing the time bucket for aggregated news signals.
- ◆ **news_count_hour**: Number of news items matching the query in that hour (news volume).
- ◆ **avg_tone_hour**: Average sentiment/tone score of the matched news in that hour (aggregated indicator provided by GDELT).
- ◆ **news_share_hour**: Relative share/visibility indicator of matched coverage in that hour.

3. Pre-integration Data Quality

Prior to integration, data quality checks on uniqueness, completeness and consistency were conducted on the two hourly datasets collected from Binance and GDELT. Key uniqueness was verified and no duplicates were found for the hourly composite key (asset, ts_hour_utc) in either source. Dataset completeness was then assessed against the expected hourly granularity over the extracted time span. As reported in the completeness table, the Binance dataset exhibits full coverage for all assets (17,520 hourly records per asset) with no missing hours, while including one additional boundary candle attributable to the inclusivity of the extraction window.

In contrast, the GDELT timeline shows slightly lower completeness (16,981 hours per asset, coverage 0.969), corresponding to 539 missing hourly buckets per asset that are not returned by the endpoint. The missing-hours figure further shows that these gaps occur at the same UTC timestamps across all cryptocurrencies, supporting the interpretation that missingness is driven by endpoint-level availability rather than asset-specific query issues.

In addition consistency checks were performed to ensure that variables fall within plausible domains and remain internally coherent. For Binance, no anomalies were detected in basic domain constraints: prices are strictly positive, the intrahour constraint $\text{high} \geq \text{low}$ always holds, and no negative trading volumes are present. For GDELT, sentiment consistency rules were enforced by treating hourly tone as not defined when **news_count_hour** = 0, and extreme tone values were filtered using an absolute-value threshold to remove implausible outliers.

dataset	Asset	Expected hours	Present hours	Missing hours	Coverage
<i>Binance</i>	BTC	17,521	17,521	0	1
<i>Binance</i>	BNB	17,521	17,521	0	1
<i>Binance</i>	SOL	17,521	17,521	0	1
<i>Binance</i>	XRP	17,521	17,521	0	1
<i>Binance</i>	ETH	17,521	17,521	0	1
<i>GDELT</i>	BTC	17520	16,981	539	0.969
<i>GDELT</i>	BNB	17520	16,981	539	0.969
<i>GDELT</i>	SOL	17520	16,981	539	0.969
<i>GDELT</i>	XRP	17520	16,981	539	0.969
<i>GDELT</i>	ETH	17520	16,981	539	0.969

Table 1 - completeness data source

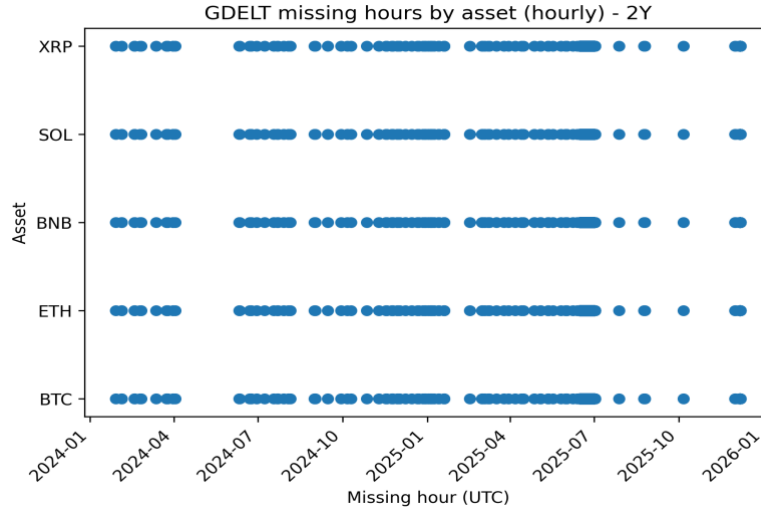


Figure 1- missing hours by asset

4. Data integration

Binance market data and GDELT news indicators were integrated at an hourly frequency by aligning both sources on a shared time key and consistent asset identifiers. Timestamps were standardized to UTC and rounded down to the hourly bucket `ts_hour_utc`. Binance symbols were standardized to a common asset identifier consistent with the news dataset, for example `BTCUSDT` was standardized to `BTC`. In both datasets, records were deduplicated on the natural key `asset, ts_hour_utc` to ensure a single observation per asset-hour and to support a one-to-one join.

The extraction window was configured consistently across both APIs and the resulting timestamp ranges were verified to ensure compatibility. Integration was implemented by defining a complete hourly reference timeline for each asset and applying left joins on the key `asset, ts_hour_utc`. Binance market variables were first left-joined onto the reference timeline, followed by a left join of GDELT indicators. This approach guarantees a regular hourly panel over the full window, while allowing GDELT values to be missing when the endpoint does not return the corresponding hour.

To distinguish true zero-news hours from missing endpoint buckets, a dedicated flag `gdeft_missing` was introduced and set when both `news_count_hour` and `avg_tone_hour` are absent after the join. Missing values in `news_count_hour` were then imputed to zero to preserve a consistent numeric series, while `gdeft_missing` retains information about potential coverage gaps.

This step completes the data integration phase by producing a unified hourly panel containing the directly observed variables from the two sources. The derivation of additional features such as returns, rolling volatility, and spike indicators is performed separately in the subsequent data enrichment phase.

5. Data Enrichment

After integration, the unified hourly panel was enriched with a set of derived features designed to support the research questions while keeping the final dataset minimal and analysis-oriented. On the market side, hourly price movement was computed from the log-transformed closing price. The one-hour log return was retained as the main price-dynamics measure and is defined as:

$$ret_{1h,t} = \ln(close_t) - \ln(close_{t-1}) = \ln\left(\frac{close_t}{close_{t-1}}\right).$$

This quantity represents the continuously-compounded return over one hour and captures short-term price variation on a comparable scale across assets with different price levels. It is central to the analysis

because it provides the basis for measuring both the magnitude of market reactions around news events and the intensity of price moves used to identify price shocks. In addition, an intrahour range indicator, `hl_range_pct`, was derived as the relative high–low range within the candle to provide an intuitive measure of within-hour price dispersion.

On the news side, the main objective was to identify abnormal surges in media attention. A binary news-spike indicator (`news_spike`) was constructed from `news_count_hour` by comparing each hour’s news volume to a high-percentile threshold (`news_p95`). Specifically, for each asset and semester, `news_p95` was defined as the 95th percentile of the hourly news-count distribution, and an hour was flagged as a spike when `news_count_hour` exceeded this threshold. Semester-specific thresholding was adopted to account for potential regime shifts in overall media intensity over a two-year horizon: when the baseline level of coverage changes across time, a single global threshold could under-detect spikes in low-coverage periods or over-detect them in high-coverage periods. Defining spikes relative to each semester therefore yields a more stable and comparable notion of “unusually high attention” within each asset’s evolving news environment. Finally, to preserve interpretability of missingness, the `gdelt_missing` flag produced during integration was kept to distinguish hours with truly zero news from hours not returned by the GDELT endpoint.

The final analysis dataset contains 87,600 rows and 16 columns, corresponding to a complete panel for five assets over two years, augmented with the selected enrichment features described above.

5.1 Post Enrichment Data Quality

After enrichment, final data quality checks were performed on the analysis-ready dataset to validate key integrity and ensure consistency prior to storage and querying. The composite key asset, `ts_hour_utc` was re-checked and no duplicates were found. As summarized in the post-enrichment completeness table, the final panel contains 87,600 rows, corresponding to a regular hourly series for each of the five assets over the two-year window. Missingness in the news source is tracked explicitly through the `gdelt_missing` flag, which allows downstream queries to distinguish true zero-news hours from endpoint-level missing buckets.

A final consistency note concerns the return-based feature: the one-hour return is computed using the previous hour closing price; therefore, `ret_1h` and derived measures such as `abs_ret_1h` are expected to be NULL for the first available hour of each asset in the dataset. This behaviour is inherent to the definition of lagged returns and does not indicate a data quality issue.

Asset	Expected hours	Present hours	Missing hours	Coverage
BTC	17,520	17,520	0	1
BNB	17,520	17,520	0	1
SOL	17,520	17,520	0	1
XRP	17,520	17,520	0	1
ETH	17,520	17,520	0	1

Table 2 - Enriched dataset completeness

6. Data Storage

The final curated dataset was stored in a relational database and queried using SQL. A relational model is well suited for this project because the data follow a fixed schema, can be naturally identified by a composite key, and must support analytical queries based on filtering, grouping, and time-window logic. In a financial-data context, integrity constraints and transactional guarantees help preserve consistency during loading and ensure that the analysis is performed on a stable and reproducible dataset.

The stored table contains one record for each asset–timestamp pair and enforces the natural composite key asset, `ts_hour_utc` as a primary key to prevent duplicates and guarantee one observation per time bucket. This structure keeps all variables required by the analyses in a single table, so the research-question queries can be executed directly without additional joins. Missingness from the news

source is retained through the `gdelt_missing` indicator, allowing downstream queries to distinguish between true zero-news hours and buckets not returned by the endpoint.

7. Data Analysis

All analyses were executed in SQL on the final table `enriched_hourly_2y`, which stores one observation per asset and timestamp. The four research questions are addressed using a consistent time-window approach based on SQL window functions. For each asset, lead and lag horizons of 1 hour, 6 hours, and 24 hours are computed by ordering observations over `ts_hour_utc` and applying rolling frames on the subsequent or previous rows. Evaluating multiple horizons is important because media activity is not necessarily instantaneous: its effects may emerge with a delay and may persist over the following hours, so the analysis explicitly tests short, medium, and day-ahead impacts.

Analyses are conducted by comparing outcomes across event conditions defined in the dataset, such as `news_spike` and `price_spike`. When future windows are evaluated, completeness of the horizon is enforced by requiring the full set of observations to be available in the window, for example by filtering to cases where the 24-hour forward window contains all expected rows. For news-related outcomes, observations are restricted to hours where the news series is available by using `gdelt_missing = 0`, ensuring that endpoint-level gaps do not affect the computed aggregates and probabilities.

To improve transparency and reproducibility, a brief SQL snippet is reported for each research question to highlight the key window-function steps used to implement the described logic.

7.1 RQ1. News spikes and subsequent price movement

Research question. Do hours characterized by a spike in news coverage coincide with larger price movements in the following horizons: 1 hour, 6 hours, and 24 hours?

Logic. The comparison is performed at the asset-hour level by splitting observations into spike and non-spike hours and then computing the average *future* absolute price movement for each group. Future movement is operationalized as the cumulative sum of `|ret_1h|` over forward horizons of 1, 6, and 24 hours. To ensure horizons are computed along the same cryptocurrency time series, window functions are applied within each asset, via `PARTITION BY asset`, so that each forward aggregate only includes subsequent hourly returns of that asset.

```
SELECT asset, ts_hour_utc, news_spike,
       SUM(ABS(ret_1h)) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
       ) AS fwd_move_24h,
       COUNT(ret_1h) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
       ) AS n24
FROM enriched_hourly_2y
WHERE gdelt_missing = 0;
-- ...
```

Results. Across all five assets, hours flagged as `news_spike=1` are followed by higher average absolute price movement than non-spike hours (`news_spike=0`) at all horizons (1h, 6h, 24h). The effect is strongest at short horizons: the 1-hour forward movement increases by +24.2% for BTC and +25.5% for XRP, while ETH and SOL show +20.3% and +17.3%, respectively; BNB exhibits a smaller but consistent uplift (+9.3%). At longer horizons the difference remains positive but tends to attenuate, ranging from +6.0% to +17.5% at 6 hours and from +1.8% to +12.5% at 24 hours (lowest for ETH, highest for XRP). This pattern indicates that news spikes are associated with persistently higher subsequent price

movement, with the largest relative impact observed within the first hours after the spike. As shown in Figure 2, results are summarized in terms of percentage uplift, computed as $\left(\frac{\text{avg_move_spike}}{\text{avg_move_non-spike}} - 1\right) \cdot 100$. The chart therefore reports the percentage increase in subsequent absolute price movement during news-spike hours relative to non-spike hours, for each asset and forward horizon.

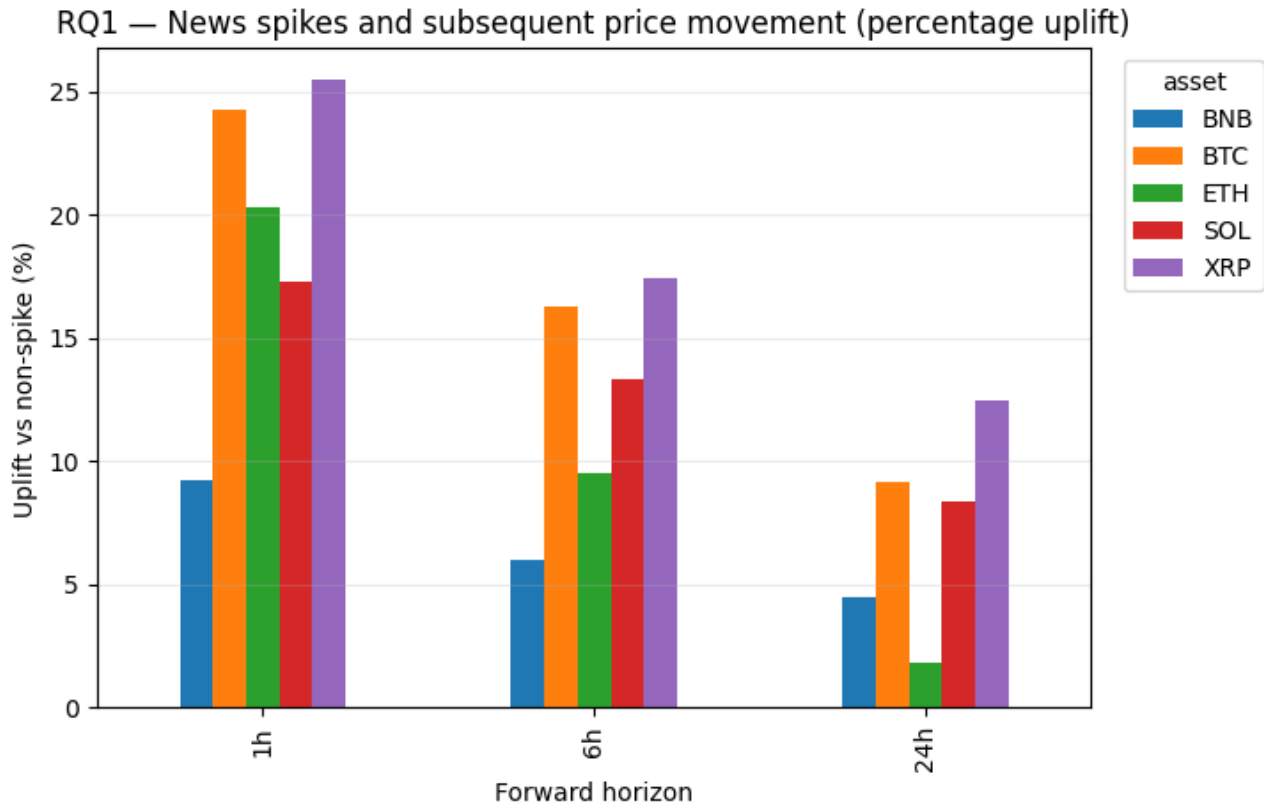


Figure 2 - RQ1 results

To provide a clearer view of the spike–non-spike contrast, Table 3 reports the mean outcomes separately for `news_spike = 1` and `news_spike = 0`.

asset	news_spike	avg_move_1h	avg_move_6h	avg_move_24h
BNB	1	0.00445	0.02594	0.10239
BNB	0	0.00407	0.02447	0.09796
BTC	1	0.00412	0.02322	0.08753
BTC	0	0.00331	0.01997	0.08018
ETH	1	0.00563	0.03094	0.11554
ETH	0	0.00468	0.02825	0.11348
SOL	1	0.00743	0.04318	0.16559
SOL	0	0.00633	0.03809	0.15281
XRP	1	0.00696	0.03944	0.15185
XRP	0	0.00555	0.03357	0.13501

Table 3 – RQ1 results

Consistent with the uplift patterns, the largest spike–non-spike gaps at short horizons are observed for XRP and BTC, with `avg_move_1h` equal to 0.0070 vs 0.0055 for XRP and 0.0041 vs 0.0033 for BTC, and `avg_move_6h` equal to 0.0394 vs 0.0336 for XRP and 0.0232 vs 0.0200 for BTC. At 24h, the divergence

remains most evident for XRP 0.1518 vs 0.1350 and SOL 0.1656 vs 0.1528, while ETH shows the smallest separation 0.1155 vs 0.1135.

7.2 RQ2. News spikes and previous price movements

Research question. Are news spikes preceded by higher price volatility in the previous horizons (same hour, 1 hour, 6 hours, 24 hours), suggesting that news coverage may react to ongoing market moves?

Logic. Using the same within-asset windowing approach introduced in RQ1, this analysis focuses on *pre-spike* dynamics. For each asset-hour, absolute movement is evaluated both contemporaneously at time t (the spike hour) and over backward-looking windows of 1, 6, and 24 hours, obtained by aggregating $|\text{ret_1h}|$ across the corresponding lookback horizons. To avoid distorted comparisons driven by incomplete windows, the sample is restricted to hours for which the full 24-hour lookback is available.

```
SELECT asset, ts_hour_utc, news_spike,
       ABS(ret_1h) AS move_same_hour,
       SUM(ABS(ret_1h)) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 24 PRECEDING AND 1 PRECEDING
       ) AS prev_move_24h,
       COUNT(ret_1h) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 24 PRECEDING AND 1 PRECEDING
       ) AS n24_prev
FROM enriched_hourly_2y
WHERE gdelt_missing = 0;
-- ...
```

Results. Across all five assets, hours classified as `news_spike = 1` are associated with higher contemporaneous and prior absolute price movement than non-spike hours. The effect is strongest for BTC and XRP. For BTC, movement is higher by +23.6% in the spike hour and +26.8% in the previous 1-hour window, and it remains positive over longer lookbacks, reaching +18.6% over the previous 6 hours and +14.8% over the previous 24 hours. XRP shows a similarly persistent pattern, with uplifts of +21.8% in the spike hour, +22.7% over the previous 1 hour, +23.9% over the previous 6 hours, and +19.3% over the previous 24 hours. SOL displays consistent increases across horizons, generally in the +14% to +20% range, while BNB shows smaller but still positive uplifts, around +8% to +11%. ETH exhibits a clear short-horizon effect, +20.6% in the spike hour and +14.6% over the previous 1 hour, while the difference largely fades at longer lookbacks, staying close to +3% over both the previous 6 and 24 hours. Overall, the pattern supports the interpretation that news spikes tend to occur during or after periods of elevated volatility, consistent with media attention reacting to recent market moves. Figure 3 summarizes these effects as the percentage uplift in average absolute movement for spike hours relative to non-spike hours.

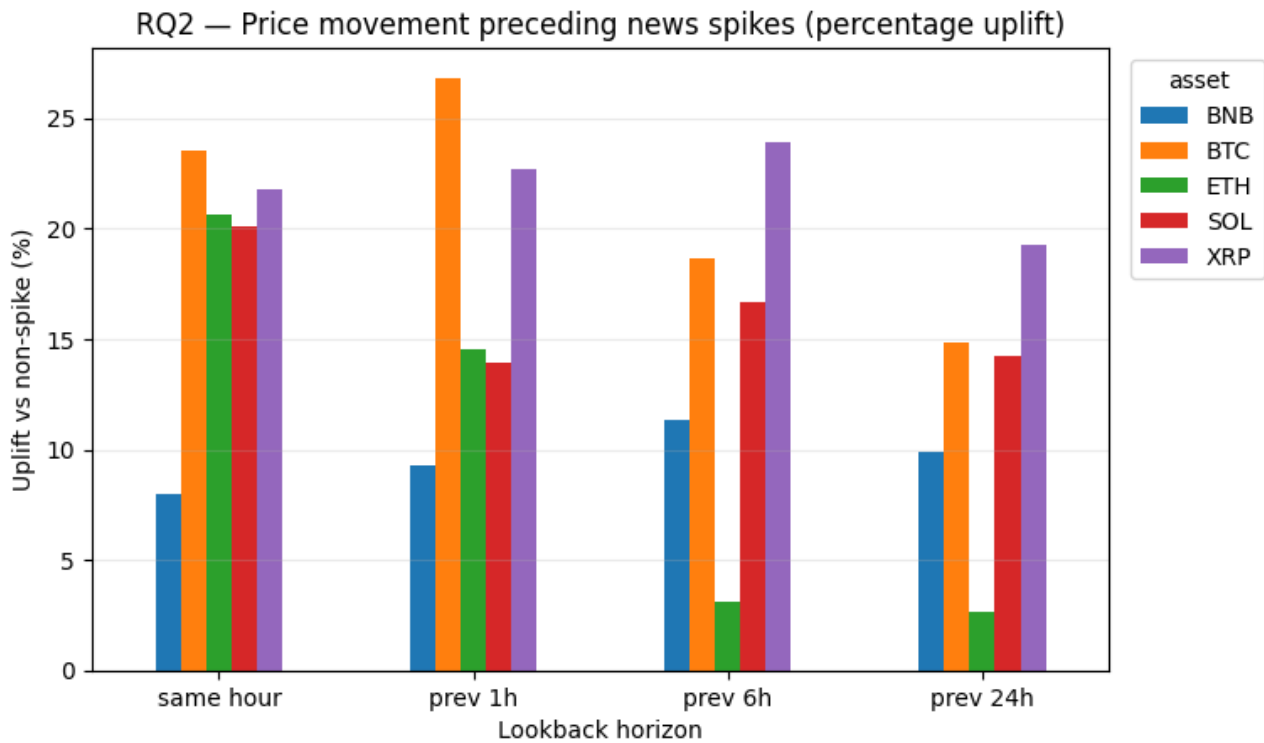


Figure 3 - RQ2 results

For completeness, Table 4 shows the underlying average outcome levels for spike and non-spike hours, comparing news_spike = 1 versus news_spike = 0.

asset	news_spike	avg_move_t	avg_prev_move_1h	avg_prev_move_6h	avg_prev_move_24h
BNB	1	0.00440	0.00444	0.02712	0.10722
BNB	0	0.00407	0.00407	0.02435	0.09755
BTC	1	0.00410	0.00420	0.02366	0.09178
BTC	0	0.00332	0.00331	0.01995	0.07994
ETH	1	0.00565	0.00538	0.02923	0.11647
ETH	0	0.00468	0.00470	0.02835	0.11342
SOL	1	0.00759	0.00723	0.04433	0.17390
SOL	0	0.00632	0.00634	0.03798	0.15223
XRP	1	0.00679	0.00683	0.04133	0.15990
XRP	0	0.00557	0.00557	0.03336	0.13405

Table 4 - RQ2 results

In line with the uplift evidence, the strongest pre-spike build-up is concentrated in XRP and SOL over longer windows, with avg_prev_move_6h at 0.0413 vs 0.0334 for XRP and 0.0443 vs 0.0380 for SOL, and avg_prev_move_24h at 0.1599 vs 0.1341 for XRP and 0.1739 vs 0.1522 for SOL. At the 1h pre-window, BTC displays the clearest gap, 0.0042 vs 0.0033, whereas ETH remains comparatively flat as the window extends, with 0.0292 vs 0.0284 at 6h and 0.1165 vs 0.1134 at 24h.

7.3 RQ3. News spikes and subsequent market activity

Research question. Do news spikes correspond to increased market activity, measured through the number of trades, in the subsequent horizons: 1 hour, 6 hours, and 24 hours?

Logic. Following the same event-style comparison framework described in RQ1, the outcome variable is switched from volatility to market activity. Trading intensity is proxied by `n_trades` and aggregated over forward windows of 1, 6, and 24 hours to quantify how activity evolves after the current hour. Average forward activity is then contrasted across spike versus non-spike hours for each asset.

```
SELECT asset, ts_hour_utc, news_spike,
       SUM(n_trades) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
       ) AS fwd_trades_24h,
       COUNT(n_trades) OVER (
         PARTITION BY asset ORDER BY ts_hour_utc
         ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
       ) AS n24
FROM enriched_hourly_2y
WHERE gdelt_missing = 0;
-- ...
```

Results. News-spike hours are followed by higher average trading activity at short horizons for all assets. In the following hour, the uplift is largest for SOL and XRP, around +38% and +37%, followed by BNB at about +35% and BTC at about +27%, while ETH shows a smaller increase of about +14%. Over the 6-hour horizon, the uplift remains sizable for BNB, SOL, and XRP, close to +28%, is moderate for BTC at about +16%, and is small for ETH at about +4%. Over 24 hours the effect attenuates but stays positive for four assets: BNB is around +24%, XRP around +20%, SOL around +18%, and BTC around +8%. ETH is the only case showing a negative 24-hour effect, about -8%. Figure 4 reports the percentage uplift in average future `n_trades` for news_spike hours relative to non-spike hours.

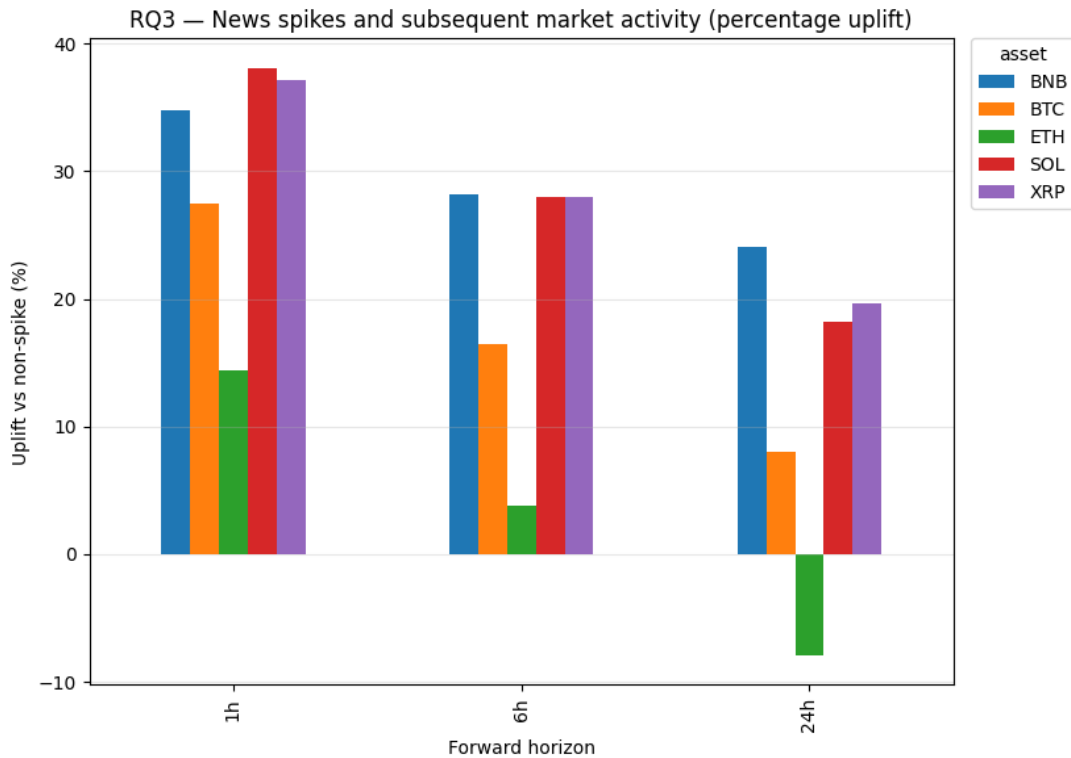


Figure 4 - RQ3 results

To better visualize the difference between spike and non-spike periods, Table 5 reports the average outcomes for `news_spike = 1` and `news_spike = 0`.

asset	news_spike	avg_fwd_trades_1h	avg_fwd_trades_6h	avg_fwd_trades_24h
BNB	1	51,742	296,744	1,152,511
BNB	0	38,385	231,515	929,275
BTC	1	174,377	961,852	3,585,670
BTC	0	136,805	825,762	3,319,955
ETH	1	143,136	784,234	2,801,398
ETH	0	125,131	755,189	3,042,853
SOL	1	106,408	595,732	2,214,070
SOL	0	77,057	465,364	1,873,632
XRP	1	71,165	402,244	1,517,881
XRP	0	51,885	314,291	1,268,381

Table 5 - RQ3 results

Echoing the uplift results, the most pronounced post-spike increases in trading activity are concentrated at 1h–6h for SOL, XRP and BNB, with avg_fwd_trades_1h equal to 106.4k vs 77.1k for SOL, 71.2k vs 51.9k for XRP and 51.7k vs 38.4k for BNB, and avg_fwd_trades_6h equal to 595.7k vs 465.4k for SOL and 402.2k vs 314.3k for XRP. Over 24h the gap remains positive for SOL 2.21M vs 1.87M, XRP 1.52M vs 1.27M and BNB 1.15M vs 0.93M, while ETH is the main exception with a reversal at 24h, 2.80M vs 3.04M.

7.4 RQ4. Price spikes and subsequent news coverage

Research question. Do extreme price movements increase subsequent news coverage, measured both as future news volume and as the probability of observing at least one news spike within the next 24 hours?

Logic. Extreme price movements are identified by ranking hourly absolute returns within each asset and semester and labeling as price_spike the observations in the top ventile. Starting from these “price-spike hours”, subsequent news coverage is quantified in two complementary ways: cumulative news volume over the next 1h, 6h, and 24h, and an indicator capturing whether at least one news spike occurs in the following 24 hours. The comparison is then carried out by contrasting these forward-looking news measures between price-spike hours and the remaining hours.

```
WITH ranked AS (
  SELECT *, NTILE(20) OVER (PARTITION BY asset ORDER BY abs_ret_1h) AS vtile
  FROM enriched_hourly_2y
  WHERE gdelt_missing = 0 AND abs_ret_1h IS NOT NULL
)
SELECT asset, ts_hour_utc,
  CASE WHEN vtile = 20 THEN 1 ELSE 0 END AS price_spike,
  SUM(news_count_hour) OVER (
    PARTITION BY asset ORDER BY ts_hour_utc
    ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
  ) AS fwd_news_24h,
  MAX(news_spike) OVER (
    PARTITION BY asset ORDER BY ts_hour_utc
    ROWS BETWEEN 1 FOLLOWING AND 24 FOLLOWING
  ) AS any_news_spike_24h
FROM ranked;
-- ...
```

Results. Price-spike hours are followed by higher average news volume for four assets out of five, with the strongest effects at short horizons. In the following hour, the uplift is largest for BTC at about +34% and XRP at about +29%, followed by BNB at about +21% and SOL at about +18%, while ETH shows a negative short-run effect of about -7%. Over the 6-hour horizon, the uplift remains substantial for BTC

and XRP, around +30% and +28%, and is around +23% for SOL and +23% for BNB, whereas ETH remains negative at about -13%. Over 24 hours, the effect attenuates but stays positive for BNB and BTC at about +21%, for SOL at about +19%, and for XRP at about +17%, while ETH shows the largest negative 24-hour effect of about -18%. The probability of observing at least one news spike within the next 24 hours increases for all assets, with the largest uplift for BTC at about +11% and SOL at about +10%, followed by XRP and BNB around +9% and +8%, while ETH shows a smaller increase of about +4%. Figure 5 reports the percentage uplift in average future news coverage for price-spike hours relative to non-spike hours.

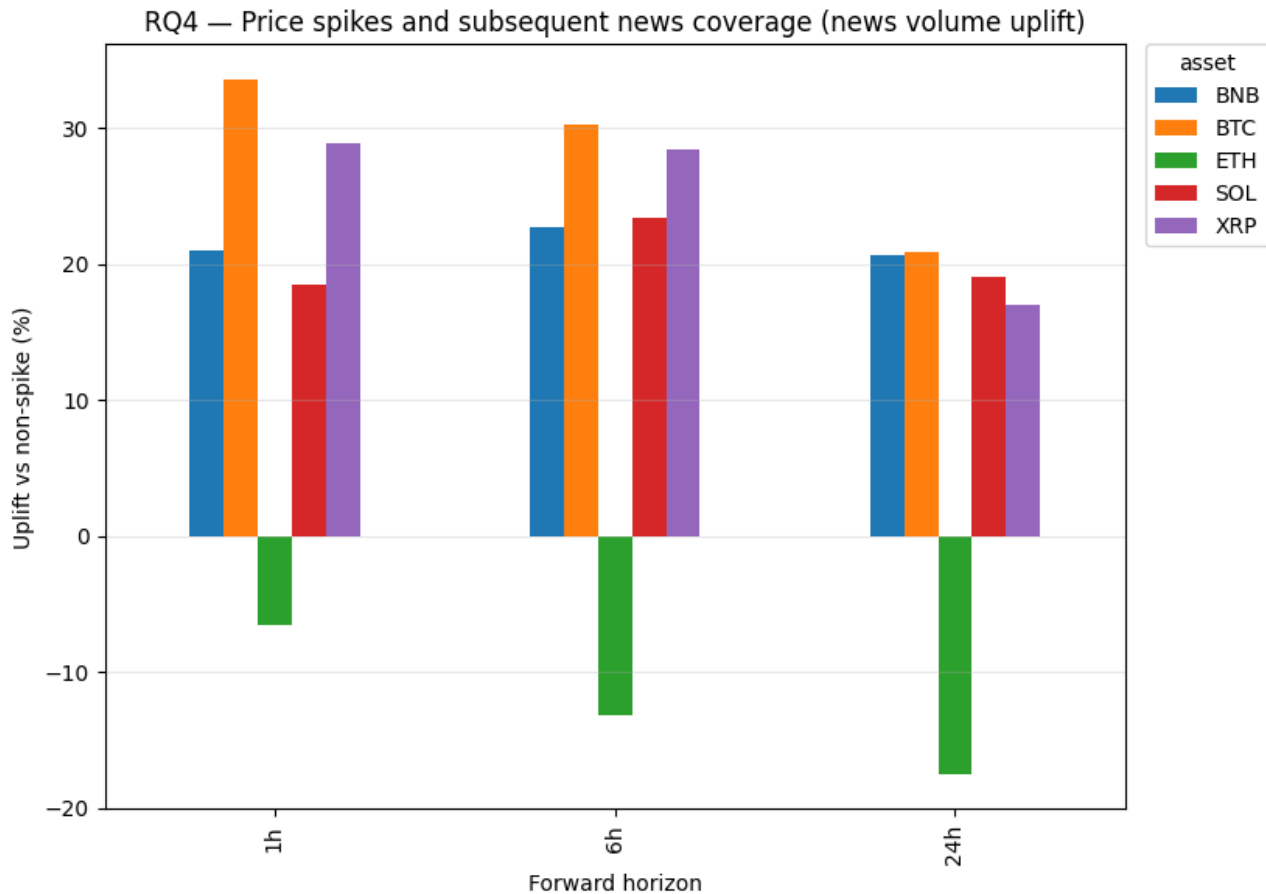


Figure 5 - RQ4 results

To complement the uplift-based comparison, Table 6 reports the average outcomes for price_spike = 1 and price_spike = 0.

asset	price_spike	avg_fwd_news_1h	avg_fwd_news_6h	avg_fwd_news_24h	p_any_news_spike_24h
BNB	1	1.327	8.065	31.731	0.740
BNB	0	1.097	6.573	26.306	0.684
BTC	1	15.904	93.144	347.418	0.517
BTC	0	11.905	71.548	287.401	0.465
ETH	1	8.025	44.852	170.857	0.528
ETH	0	8.585	51.684	207.130	0.506
SOL	1	1.588	9.903	38.309	0.777
SOL	0	1.341	8.027	32.171	0.706
XRP	1	0.771	4.614	16.902	0.893
XRP	0	0.598	3.592	14.449	0.822

Table 6 - RQ4 results

Echoing the uplift results, the most pronounced post-event increases in news volume are observed for BTC and SOL at longer horizons, with `avg_fwd_news_24h` equal to 347.42 vs 287.40 for BTC and 38.31 vs 32.17 for SOL, alongside higher spike probability within 24h at 0.517 vs 0.465 and 0.777 vs 0.706, respectively. XRP and BNB follow the same pattern, with `avg_fwd_news_24h` equal to 16.90 vs 14.45 for XRP and 31.73 vs 26.31 for BNB, and `p_any_news_spike_24h` increasing from 0.822 to 0.893 and from 0.684 to 0.740. ETH remains the main exception in terms of volume, with lower future news counts at all horizons, 8.03 vs 8.59 at 1h, 44.85 vs 51.68 at 6h, and 170.86 vs 207.13 at 24h, although its probability of observing a subsequent news spike still increases slightly, 0.528 vs 0.506.

Conclusions

This project built an analysis-oriented data management pipeline to study the relationship between cryptocurrency markets and news dynamics at an hourly frequency. Market microstructure data were acquired from Binance and aligned with news indicators derived from GDELT; both sources were standardized to a common UTC time key and integrated into a regular hourly panel. Data quality checks were applied both before and after integration to ensure consistency, completeness, and a clear distinction between genuinely observed zero-news hours and missing news buckets. The resulting unified dataset was enriched with a minimal set of derived variables tailored to the research questions, including log-returns, spike indicators defined through distributional thresholds, and forward/backward window aggregates computed coherently along each asset time series.

The analysis highlights a clear and consistent short-horizon pattern. When a news spike occurs, subsequent absolute price movement is higher for all assets, with the strongest relative increases typically appearing within the first hour, especially for BTC and XRP, while BNB shows a smaller but stable uplift and the effect tends to fade over 24 hours, most notably for ETH. The backward-looking results are coherent with this picture, showing that spike hours usually occur during periods of already elevated movement, again most clearly for BTC and XRP, whereas ETH mainly exhibits this difference only at very short lookbacks. Market activity reacts even more sharply than prices: trading intensity jumps markedly after news spikes at 1h–6h for SOL, XRP, and BNB, remains positive for most assets over 24 hours, and reverses only for ETH at the daily horizon. Finally, the reverse direction is also supported: extreme price movements are followed by higher subsequent news volume and a higher probability of observing a news spike within 24 hours for BTC, XRP, SOL, and BNB, while ETH stands out as the main exception, showing a decrease in future news volume despite a slightly higher spike probability. Overall, the largest discrepancies concentrate in the first hours after the event and are most pronounced for BTC, XRP, SOL, and BNB, with ETH consistently exhibiting the weakest or most mixed effects at longer horizons.

Limitations

This study is primarily descriptive and observational. It compares market and news outcomes across spike and non-spike hours, but it does not aim to identify causal effects. Cryptocurrency prices and trading activity are shaped by many drivers beyond news coverage, including broader market conditions and macro events. For this reason, the results should be interpreted as empirical associations that summarize recurring patterns in the data rather than as evidence of direct causal mechanisms.

In addition, the news indicators are derived from GDELT through query-based filtering and aggregation. While this approach provides a scalable proxy for media pressure, it cannot guarantee perfect coverage of all relevant crypto-related articles. Some relevant news may not match the query design and therefore be missed, while some retrieved items may only partially relate to the target assets, generating false positives.

Finally, the analysis relies on fixed evaluation horizons of 1h, 6h, and 24h. This choice improves interpretability and comparability across research questions, but it also simplifies the timing of potential reactions. In practice, the speed and persistence of market and media responses can vary across assets, periods, and event types, and relevant effects may occur at shorter or longer lags than the selected windows.

Future work

Future work could extend this study by enriching the dataset with additional variables that capture broader market forces and contextual drivers of price dynamics. Since crypto volatility reflects more than media attention alone, integrating further market-related information would support a deeper interpretation of when and why the news–market relationship becomes stronger or weaker.

Another natural step is to move from descriptive comparisons to a more explicitly statistical analysis based, for example, on modeling approaches. This would allow a deeper investigation of price dynamics and trading activity, and it would also provide a more flexible representation of delayed responses. Instead of assuming that the relevant delay must be 1h, 6h, or 24h, the lag structure could be estimated from data. A practical option is to adopt adstock-type transformations, widely used, for example, in marketing mix models to represent delayed and decaying exposure effects, as a compact way to model how media pressure may propagate over time.

References

<https://github.com/public-apis/public-apis?tab=readme-ov-file#index>

BINANCE Spot REST API

- Base: <https://api.binance.com>
- GET <https://api.binance.com/api/v3/klines>
- GET <https://api.binance.com/api/v3/exchangeInfo>

GDELT DOC 2.0 API

- Base/endpoint: <https://api.gdeltproject.org/api/v2/doc/doc>
- (mode usati: timelinevolraw, timelinetone, timelinevol)

Meta. Robyn. URL <https://facebookexperimental.github.io/Robyn/>.

Dr. R. Jarrar, Mass Analytcs. Your Comprehensive Marketing Mix Modeling Guide.